Using Multi-omics Integration as a Model for Cancer Research

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1 Introduction

Recent advancements in high-throughput technologies have led to the generation of vast amounts of molecular data, opening new avenues for exploring the mechanisms underlying cellular state changes. Traditionally, cellular states are inferred from a single type of biomolecule—such as RNA expression measured by RNA-seq. While these approaches have advanced our understanding of cell dynamics, analyzing RNA data in isolation fails to fully capture the complexity of cellular states. To overcome this limitation, integrating multiple omics layers—genomic, transcriptomic, and proteomic—is essential for obtaining a more comprehensive view of biological processes. This is particularly relevant in cancer research, where understanding tumor heterogeneity is critical for improving diagnosis and treatment. To this end, we introduce MOIE (Multi-Omics Integration Ensemble), a two-tiered ensemble learning framework that integrates multi-omics data to enhance cancer subtype and stage classification while identifying biologically meaningful biomarkers.

2 Background

Multi-omics integration has become a powerful approach in cancer research, enabling a more comprehensive understanding of tumor heterogeneity and underlying biological processes. Ensemble modeling—a machine learning technique that combines predictions from multiple base learners—has shown promise in integrating heterogeneous omics datasets for improved classification and prediction.

Several tools have contributed to the advancement of multi-omics analysis. For example, mixOmics enables feature selection and integration across omics layers using multivariate statistics; DeepProg combines deep learning with boosting techniques for survival prediction; and MoGCN applies graph convolutional networks to learn relationships across omics types for cancer subtype classification. However, these methods face limitations. Some rely heavily on survival information, are constrained to single-outcome predictions, or lack flexibility in handling modality imbalance.

By building on the strengths of existing frameworks while addressing their weaknesses, this study proposes a new ensemble-based strategy for integrating multi-omics data. The goal is to improve predictive performance, enhance biomarker discovery, and support more effective computational tools for cancer subtype and stage classification.

3 Materials and Methods

3.1 Datasets

This study utilizes multi-omics data from two cancer types in The Cancer Genome Atlas (TCGA): breast invasive carcinoma (BRCA) and the pan-kidney cohort (KIPAN). Preprocessed molecular data and associated clinical information were obtained from the LinkedOmics portal, a web application that aggregates data from TCGA and CPTAC cancer cohorts. These datasets were selected for their large cohort sizes and frequent use in machine learning studies.

From the TCGA-KIPAN cohort, we constructed two datasets with distinct clinical targets. The first integrates RNA sequencing (RNASeq), copy number variation (CNV), and reverse phase protein array (RPPA) data, with cancer subtype as the classification target. RNASeq data represent normalized gene transcript expression; CNV reflects normalized copy number and alterations in segmented genomic regions; RPPA captures normalized protein expression levels. The classification includes three histological subtypes: kidney chromophobe (KICH), kidney clear cell carcinoma (KIRC), and kidney papillary cell carcinoma (KIRP). Of the 941 available samples in the KIPAN cohort, 736 had complete data across all three omics modalities.

The second KIPAN dataset integrates RNASeq, methylation, and RPPA data, with pathological stage as the clinical endpoint. Methylation values are expressed as beta scores mapped to the genome. Clinical staging includes four levels (I–IV), but due to class imbalance, stages I and II were grouped as “early” and stages III and IV as “late.” This yielded a binary classification task with 350 early-stage and 208 late-stage samples.

The third dataset, derived from the TCGA-BRCA cohort, includes RNASeq, miRNA, and methylation data. The classification task uses the four intrinsic subtypes defined by the PAM50 gene expression signature: Basal-like, HER2-enriched, Luminal A, and Luminal B. While the full cohort includes 1,098 tumor samples, 430 had complete data for all three omics layers and were used in this analysis.

3.2 Preprocessing

For each dataset, samples were first filtered to ensure complete representation across all included omics modalities, resulting in equal sample sizes for integration. To reduce dimensionality and improve data quality, features with missing values were removed.

RNA and miRNA data were normalized, log-transformed, and filtered to retain the most variable genes using `scanpy` preprocessing tools. For methylation, CNV, and RPPA datasets, low-variance features were excluded based on predefined thresholds.

To address remaining missing values, CNV data were imputed using a simple mean-based imputation strategy, while RPPA data were imputed using k-nearest neighbors (KNN) imputation.

3.3 Base estimators

This study employed a combination of linear and tree-based models as base estimators within the ensemble learning framework. The selected models include logistic regression, balanced random forest, multi-layer perceptron (MLP), and support vector classification (SVC). All of the estimators were implemented using `scikit-learn` library, with the exception of `BalancedRandomForestClassifier`, which was sourced from the `imblearn` library.

These models were chosen for their demonstrated performance on classification tasks, as well as their versatility and compatibility with high-dimensional, multi-omics datasets.

3.4 Ensemble modeling

To evaluate whether combining different base learners improves classification performance for each omics modality, we implemented a stacking-based ensemble strategy. Stacking enables the integration of heterogeneous models by learning a meta-model that combines their predictions.

We developed a custom modality-level ensemble model using `scikit-learn`'s `BaseEstimator` and `TransformerMixin` classes. For stacking functionality, we used the `stackingClassifier` from the `ensemble` module, with `logistic regression` as the final default estimator.

Each base estimator was trained on the full input dataset, while the final estimator was trained on cross-validated predictions from the base learners using the `cross_val_predict` function. This architecture leverages the strengths of individual models while mitigating their individual weaknesses, leading to improved predictive performance.

3.5 Performance

To assess model performance, we employed 5-fold stratified cross-validation with 3 repeats, implemented using `scikit-learn`'s `StratifiedKFold`. Performance was evaluated using the F1 score and balanced accuracy, calculated via the `sklearn.metrics` and `imblearn.metrics` modules, respectively.

To assess the statistical significance of observed differences in model performance, we compared results across the 15 cross-validation runs using a two-sided Wilcoxon signed-rank test. This analysis was used to determine whether the integration ensemble performed similarly or better than the modality-level ensemble. The test was implemented using the `wilcoxon` function from the `scipy.stats` module.

3.6 Feature importance

To identify genes most important to the ensemble model, we utilized the `permutation_importance` method from `scikit-learn`. In this context, permutation importance quantifies the average reduction in balanced accuracy resulting from randomly shuffling the values of each feature. A larger reduction indicates higher feature importance.

Each gene was permuted 10 times to obtain a stable estimate of its effect on model performance. The resulting importance scores were normalized by their standard deviations and ranked in descending order to prioritize the most informative features.

4 Results

4.1 Base Learners vs. Modality Ensembles

In our evaluation of classification performance across various omics datasets, the Modality Ensemble consistently outperformed or matched the top-performing base learners in terms of F1 score. For the subtyping tasks, ensemble models achieved the highest F1 scores across all modalities where it outperformed all other models. In the BRCA subtyping task, the Modality Ensemble achieved a notable performance gap between the ensemble and the best-performing base learners. For KIPAN staging, the ensemble model yielded the highest F1 scores for the Methylation modality but failed to outperform certain base estimators for the RNA and RPPA modalities.

Table 1: Mean F1 Scores of Base Learners vs. Modality Ensembles

Model	**KIPAN Subtyping**			**KIPAN Staging**		
	RNA	CNV	RPPA	RNA	Methylation	RPPA
Logistic Regression	0.958	0.859	0.864	0.736	0.697	0.687
Balanced Random Forest	0.947	0.890	0.966	**0.763**	0.743	0.712
Deep Neural Network	0.954	0.855	0.964	0.734	0.718	**0.713**
Support Vector Classifier	0.958	0.722	0.846	0.723	0.700	0.652
Modality Ensemble	**0.958**	**0.895**	**0.977**	0.750	**0.757**	0.710

Model	**BRCA Subtyping**		
	RNA	miRNA	Methylation
Logistic Regression	0.894	0.796	0.830
Balanced Random Forest	0.900	0.787	0.786
Deep Neural Network	0.884	0.774	0.818
Support Vector Classifier	0.890	0.775	0.830
Modality Ensemble	**0.925**	**0.842**	**0.830**

4.2 Modality Ensembles vs. Multi-omics Integration Ensemble

We evaluated the performance of single-modality and multi-omics ensemble models across the three classification tasks (Table 2). RNA consistently emerged as the strongest individual modality. For KIPAN Subtyping, the combination of RNA and RPPA achieved the highest performance, outperforming all single-modality and other ensemble models. Similarly, for KIPAN Staging, the best results were obtained with the RNA + RPPA ensemble, indicating the complementary value of RPPA in staging tasks. In the case of BRCA Subtyping, the RNA + miRNA ensemble yielded the top performance, slightly outperforming both single-modality and three-way ensembles.

4.3 Feature importance

Permutation importance was done only for the BRCA dataset, for which the top most important features were methylation data.

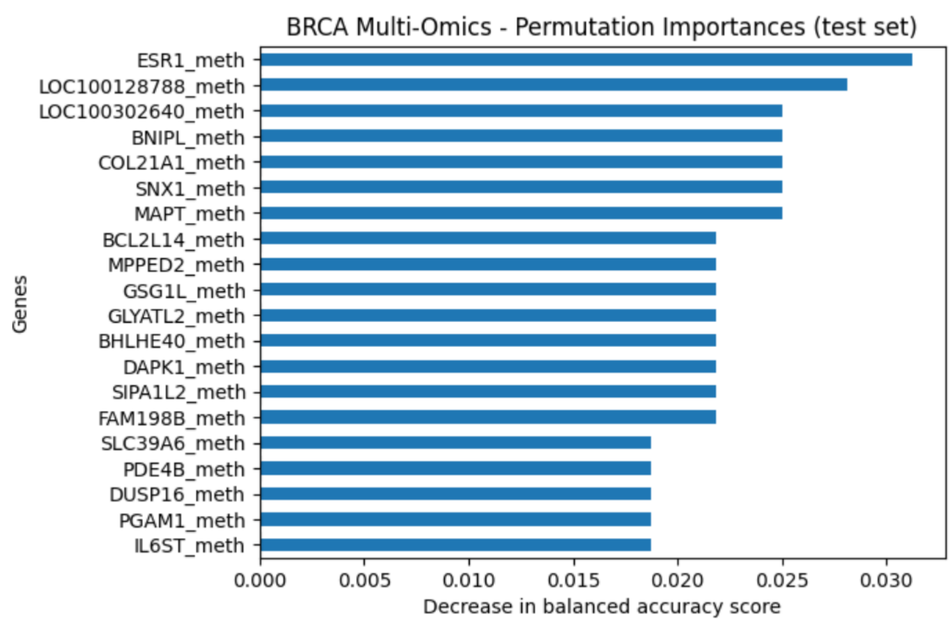

Figure 1: Top 20 most important genes to the Multi-Omics Integration Ensemble models for BRCA subtyping targets

Table 2: Mean Balanced Accuracy and F1 Scores of Modality Ensembles vs. Multi-Omics Integration Ensembles

KIPAN Subtyping		
Model	**Balanced Accuracy**	**F1 Score**
rna	0.958 +/- 0.029	0.961 +/- 0.017
cnv	0.846 +/- 0.034	0.889 +/- 0.022
rppa	0.931 +/- 0.039	0.968 +/- 0.016
rna + cnv	0.949 +/- 0.033	0.958 +/- 0.022
rna + rppa	**0.974 +/- 0.025**	**0.984 +/- 0.013**
cnv + rppa	0.965 +/- 0.024	0.943 +/- 0.066
rna + cnv + rppa	0.969 +/- 0.027	0.977 +/- 0.017
KIPAN Staging		
rna	0.718 +/- 0.037	0.746 +/- 0.034
meth	0.705 +/- 0.023	0.732 +/- 0.025
rppa	0.675 +/- 0.029	0.710 +/- 0.030
rna + meth	0.720 +/- 0.018	0.750 +/- 0.020
rna + rppa	**0.728 +/- 0.021**	**0.758 +/- 0.020**
meth + rppa	0.717 +/- 0.024	0.748 +/- 0.027
rna + meth + rppa	0.722 +/- 0.012	0.752 +/- 0.015
BRCA Subtyping		
rna	0.925 +/- 0.012	0.922 +/- 0.018
mirna	0.797 +/- 0.047	0.835 +/- 0.037
meth	0.815 +/- 0.063	0.826 +/- 0.051
rna + mirna	**0.926 +/- 0.017**	**0.926 +/- 0.014**
rna + meth	0.919 +/- 0.020	0.917 +/- 0.016
mirna + meth	0.827 +/- 0.027	0.846 +/- 0.009
rna + mirna + meth	0.925 +/- 0.023	0.924 +/- 0.015

5 Discussion

Cancer is widely recognized as a highly heterogeneous disease, and early diagnosis and prognosis remain central goals of cancer research. Multi-omics integration provides a promising avenue to better understand the complex biological mechanisms underlying cancer, yet doing so effectively and efficiently presents ongoing challenges in bioinformatics.

In this study, we developed MOIE—a multi-omics integration pipeline for cancer subtype classification—using a two-tier ensemble learning approach. Our results demonstrate the utility of ensemble methods: the use of scikit-learn's `StackingClassifier` improved predictive performance across most classification tasks by effectively combining base learners within each modality and integrating information across modalities.

For all three datasets, ensemble learning improved performance when applied to individual modalities. We also observed performance gains when combining modalities. Notably, combining lower-performing modalities resulted in improved classification performance. This suggests that ensemble models can be especially useful in contexts where high-quality datasets are unavailable, helping to extract complementary information from less informative sources.

We also explored the use of ensemble models for biomarker discovery. Using permutation-based feature importance, we identified gene sets from the BRCA dataset that were more important for model training.

Overall, this study highlights the promise of ensemble learning for both predictive modeling and biological discovery in cancer research using multi-omics data.

5.1 Limitation and further work

This study has demonstrated the effectiveness of ensemble learning in improving predictive performance for cancer classification tasks using multi-omics data. However, several limitations should be acknowledged.

First, the base learners and ensemble models were not subjected to extensive hyperparameter tuning. While this was not the primary focus of the current study, tuning could lead to further improvements in predictive performance. Future work should explore systematic optimization of model parameters to enhance accuracy and robustness.

Second, the use of permutation importance for feature evaluation is susceptible to correlation bias. In high-dimensional omics data, where gene expression features are often highly correlated, permutation importance may underestimate the relevance of individual genes. Downstream analysis can be done to verify the importance of the features selected.

Lastly, we did not compare the performance of our model against the existing multi-omics integration models such as mixOmics and MoGCN to see whether our model shows improvement.

Despite these limitations, our findings highlight the promise of ensemble learning in multi-omics integration for cancer diagnosis and biomarker discovery. Future studies can build on this framework to develop more accurate and interpretable models for cancer subtype classification.

6 Conclusion

This study demonstrates the effectiveness of ensemble machine learning methods for cancer classification using multi-omics data. Across three classification tasks, ensemble models that integrated multiple omics modalities consistently outperformed models trained on individual datasets alone. Furthermore, our use of permutation-based feature importance revealed candidate biomarkers relevant to disease subtypes and stages. These findings suggest that multi-omics integration not only improves predictive performance but also enhances biological interpretability. Future work incorporating functional enrichment analyses could further validate these biomarkers and refine their clinical utility. Overall, this study underscores the potential of ensemble learning in multi-omics frameworks for advancing cancer diagnosis and personalized treatment strategies.